

Queimadas Reference Satellite: A data analysis

Analysing data for choosing the next reference satellite for the Queimadas Program

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Agenda I

- 1 Introduction
- 2 Exploratory Data Analysis
- 3 Time Series
 - Time series decomposition
 - Reference satellites
 - Breakpoint identification
- 4 *Queimadas* method (1-month regression)
 - Regression
 - Forecast
 - Forecast differences
- 5 Future work
- 6 References



Introduction



What's new?

- Queimadas method:
 - ▶ Outliers tables.
 - ▶ Confidence intervals.
 - ▶ Use of a month-by-month regression and forecast.
 - ▶ Forecasts' differences between Queimadas 12 vs. month-by-month.
 - ▶ BFast analysis.



What would be the next reference satellite?

- What would be the next reference satellite [1, 3]?



Online resources

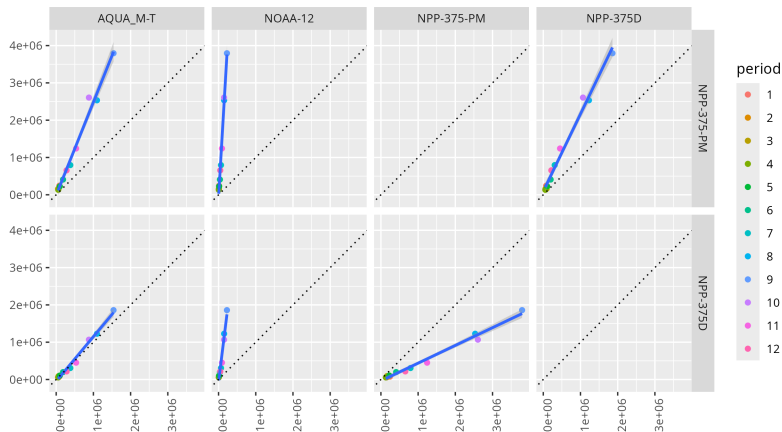
- Queimadas [portal](#).
- Queimadas [FAQ](#).
- Source code
 - ▶ Code repository available [here](#).
 - ▶ Slides available [here](#).



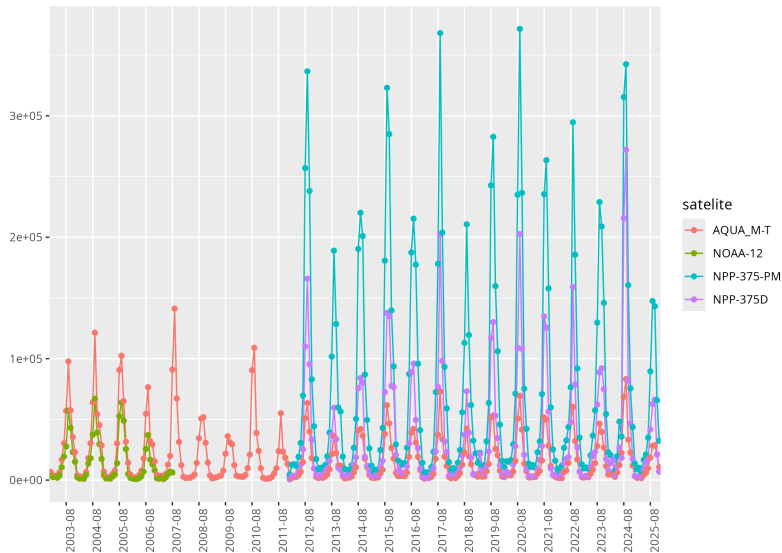
Exploratory Data Analysis



Brazilian fire foci aggregated by month (all years)

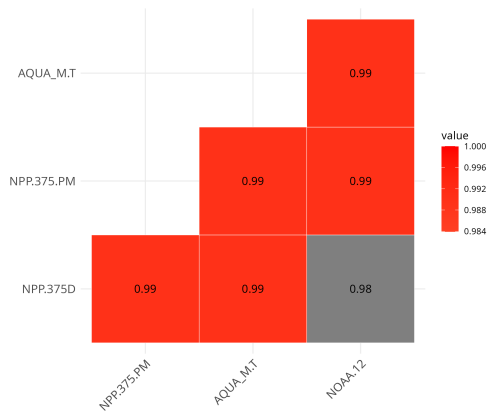


Brazilian fire foci aggregated by year and month



Correlation between fire foci from the reference satellites in Brazil

- The time series overlap partially.
- Use only the month (not the year) to compute correlation.
- This way we can use the whole time series in the estimation.

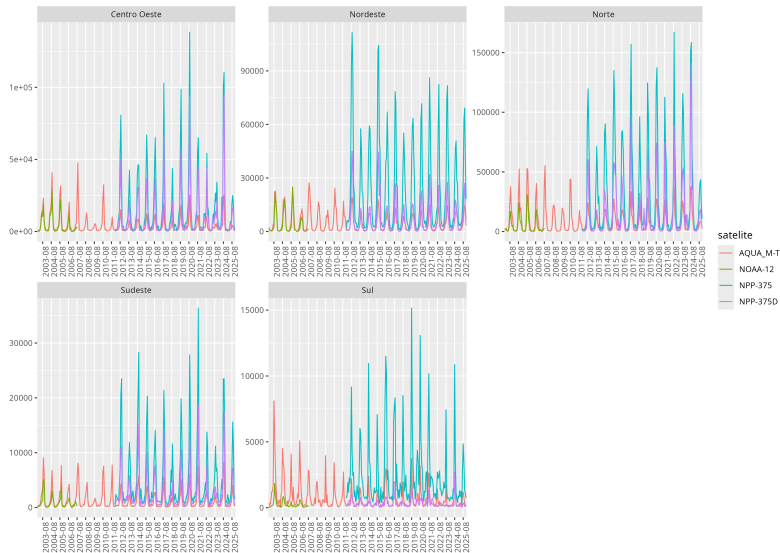


Correlation between fire foci from the reference satellites in Brazil

- The time series overlap partially.
- Estimate the correlation using only observation that overlap.
- Some satellites don't overlap and their correlation can't be estimated.

satelite_x	satelite_y	cor_val
NOAA-12	AQUA_M-T	0.922
AQUA_M-T	NPP-375-PM	0.978
AQUA_M-T	NPP-375D	0.962
NPP-375-PM	NPP-375D	0.958

Brazilian regions by year and month



Time Series

Time series' components

- Structural time series model [2]:

- ▶ Trend $g(t)$.
- ▶ Periodic changes $s(t)$.
- ▶ Effects of holidays $h(t)$.

- **Assumptions:**

- ▶ ϵ_t is normally distributed.
- ▶ Forecasting is a curve-fitting problem; no account for temporal dependence structure in the data [4].

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

Do the models fit?

- Do the data have a trend when smoothed?
- Do the data have a trend after applying the logarithm function and smoothing?
- Are the data autocorrelated in the time dimension?
- Do the data have a trend component?
- Do the data have season component?
- How does the model fit its observations?
- How do compare the residuals and the fitted values?
- Do the mode-observations residual have a normal distribution?



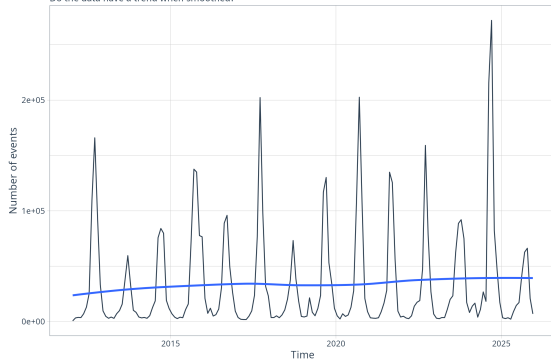
NPP-375D



Time series trend (with smoothing)

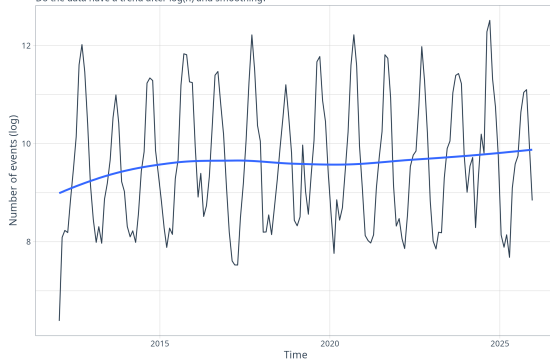
NPP-375D

Do the data have a trend when smoothed?



NPP-375D

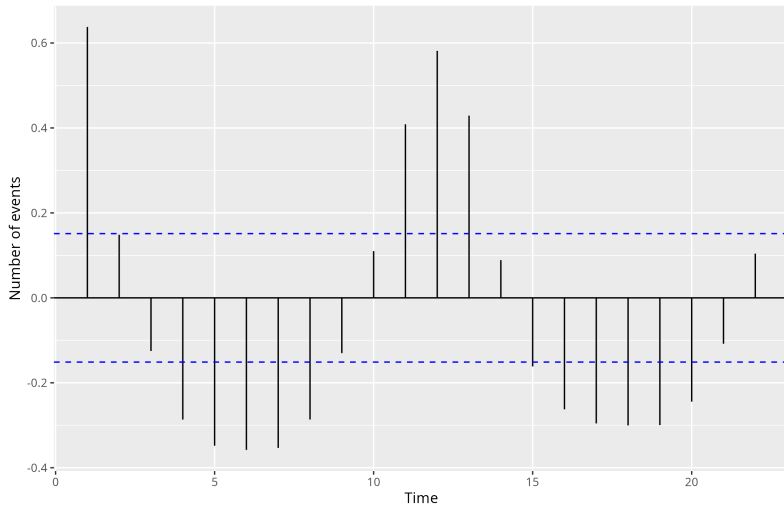
Do the data have a trend after log(n) and smoothing?



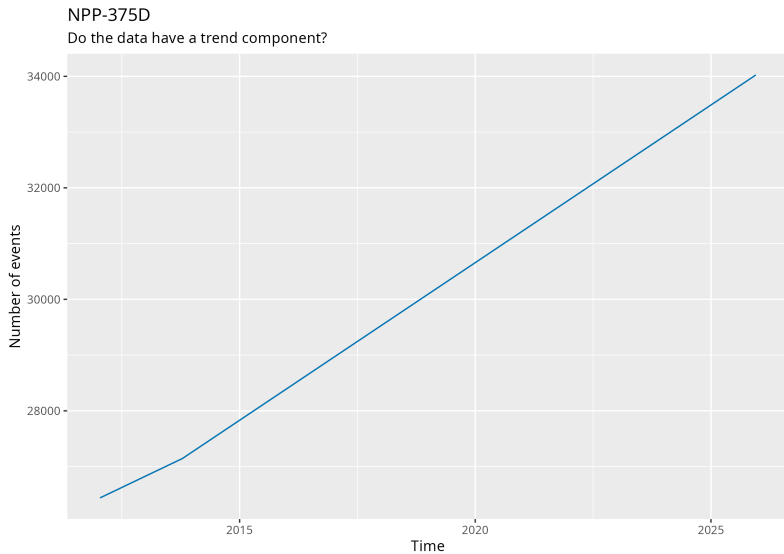
Time series autocorrelation

NPP-375D

Is the time series autocorrelated?



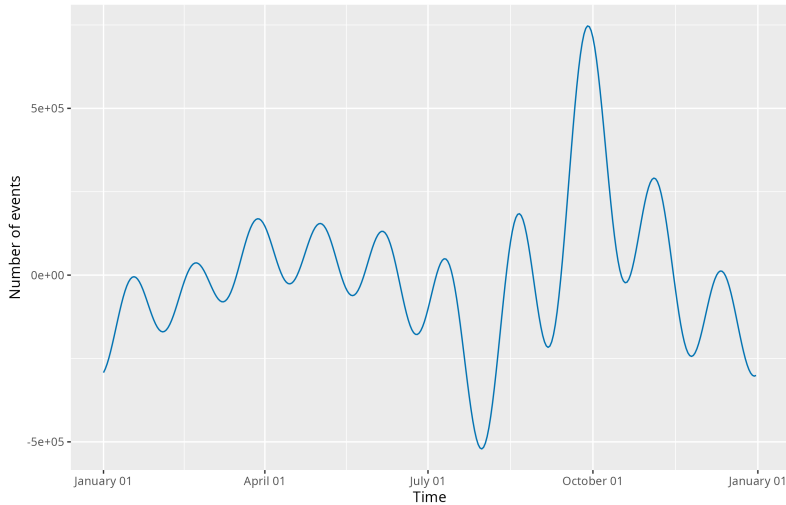
Time series trend



Time series seasonality

NPP-375D

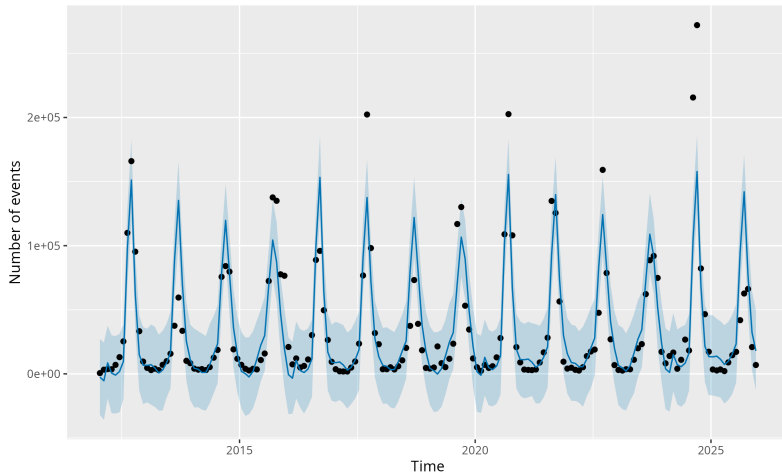
Do the data have a seasonality component?



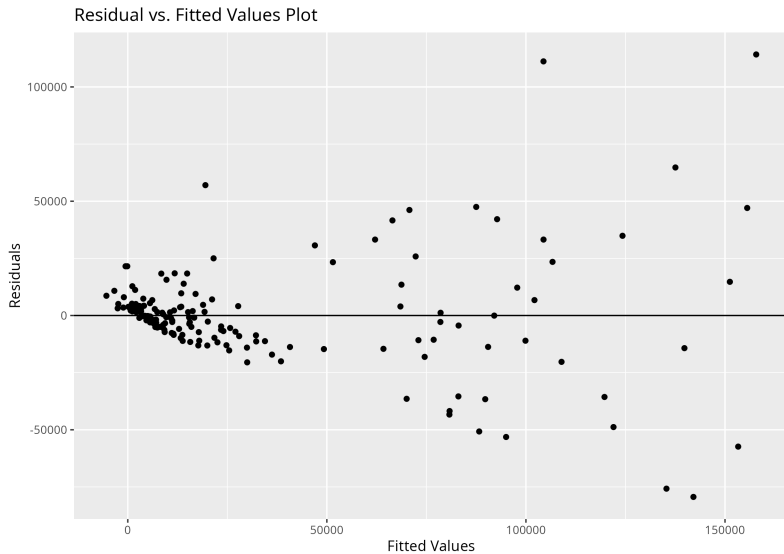
Time series versus forecast

NPP-375D

Observations (points) versus forecasting (line)



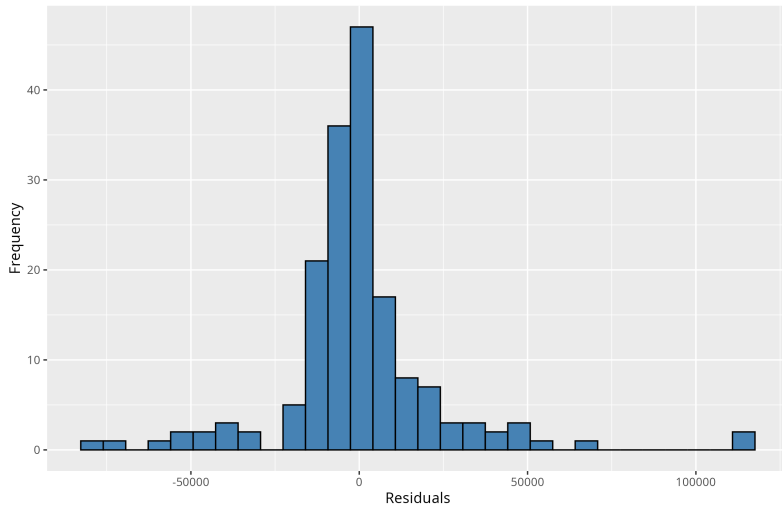
Time series versus forecast



Time series versus forecast

Histogram of Residuals

Do the residuals have a normal distribution?



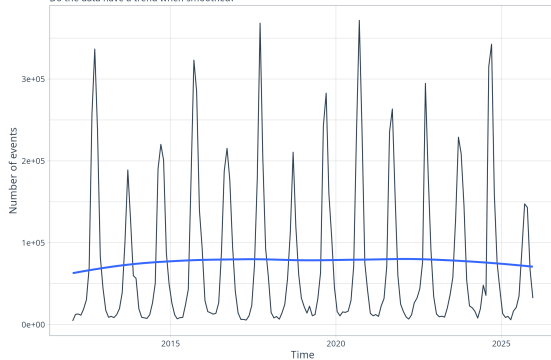
NPP-375-PM



Time series trend (with smoothing)

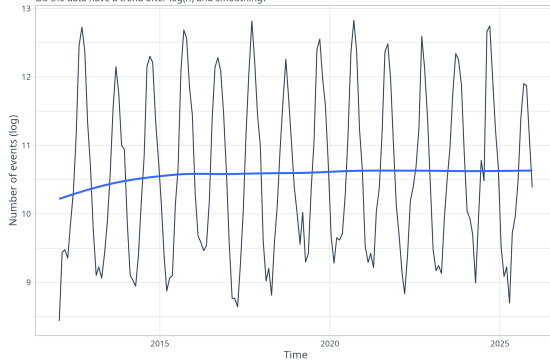
NPP-375-PM

Do the data have a trend when smoothed?



NPP-375-PM

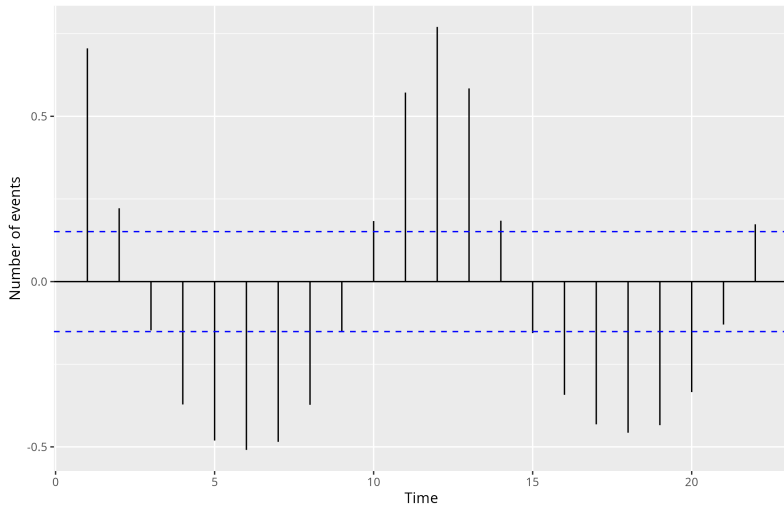
Do the data have a trend after log(n) and smoothing?



Time series autocorrelation

NPP-375-PM

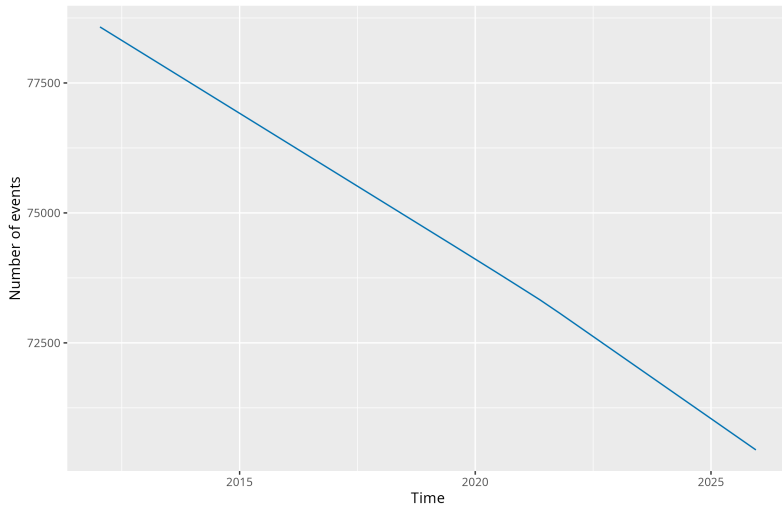
Is the time series autocorrelated?



Time series trend

NPP-375-PM

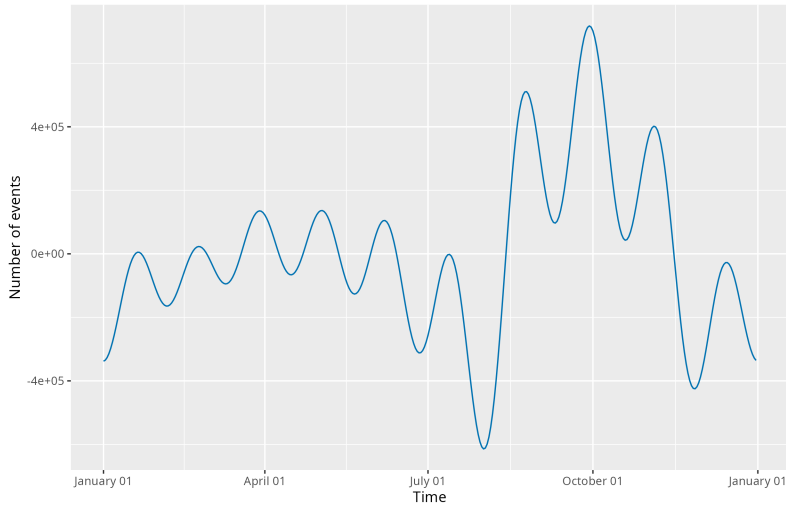
Do the data have a trend component?



Time series seasonality

NPP-375-PM

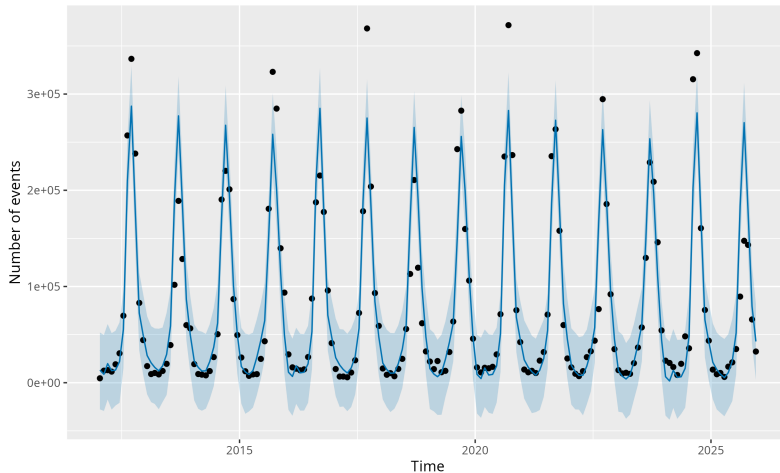
Do the data have a seasonality component?



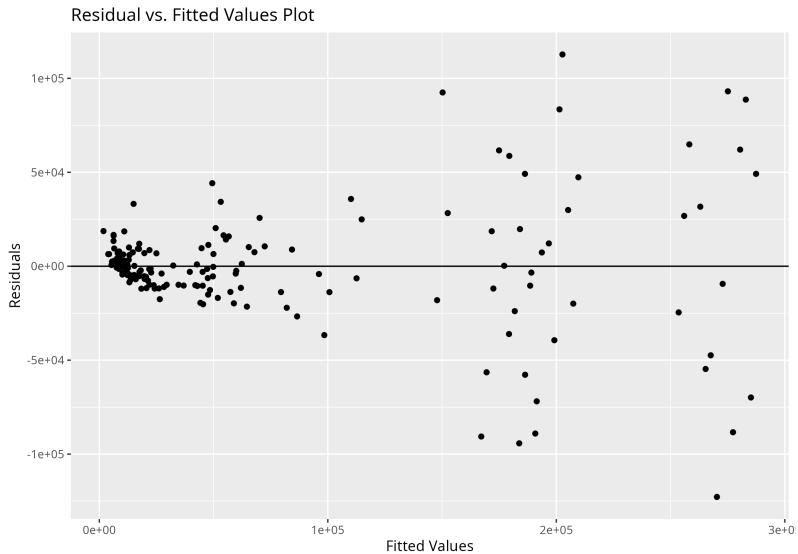
Time series versus forecast

NPP-375-PM

Observations (points) versus forecasting (line)



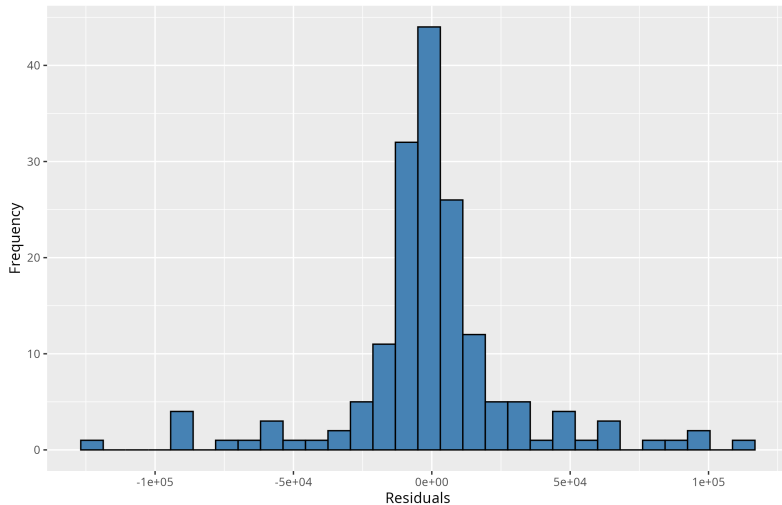
Time series versus forecast



Time series versus forecast

Histogram of Residuals

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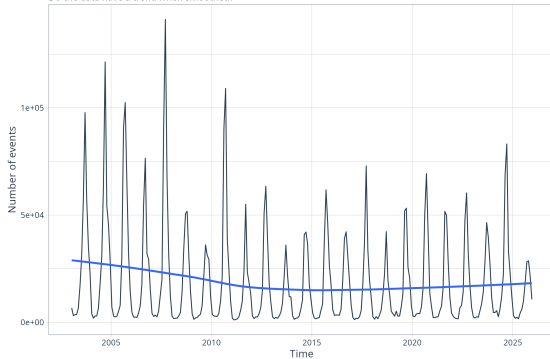
AQUA M T



Time series trend (with smoothing)

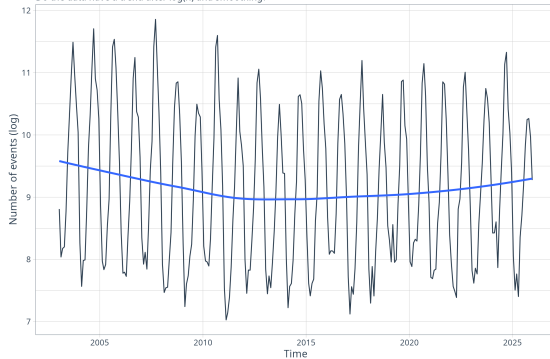
AQUA_M-T

Do the data have a trend when smoothed?

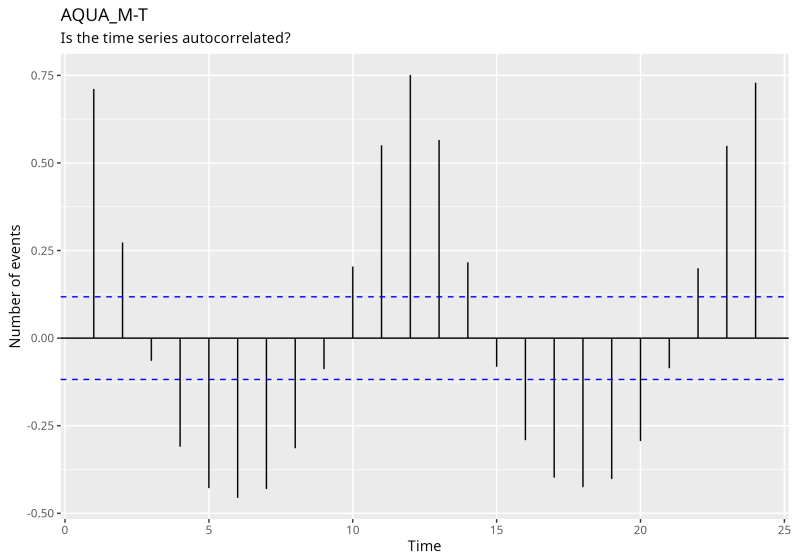


AQUA_M-T

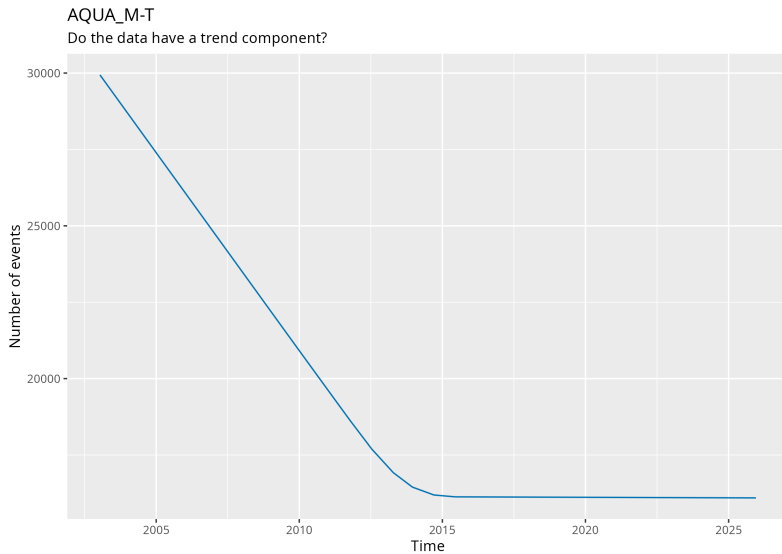
Do the data have a trend after log(n) and smoothing?



Time series autocorrelation



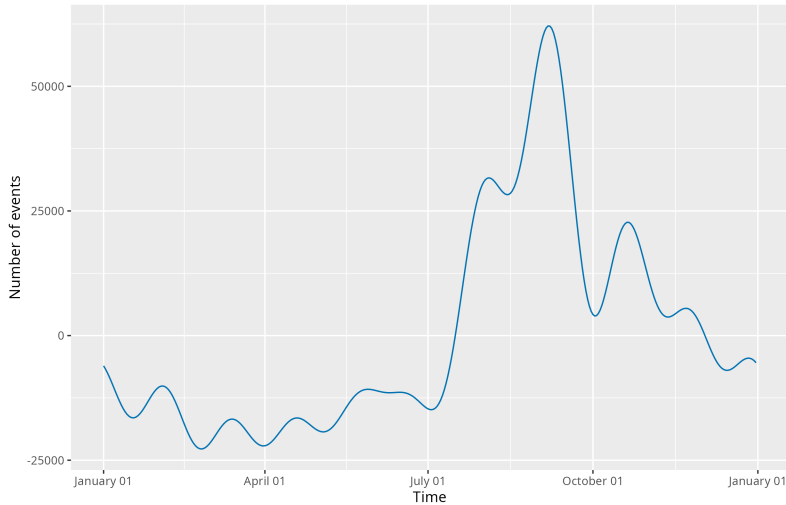
Time series trend



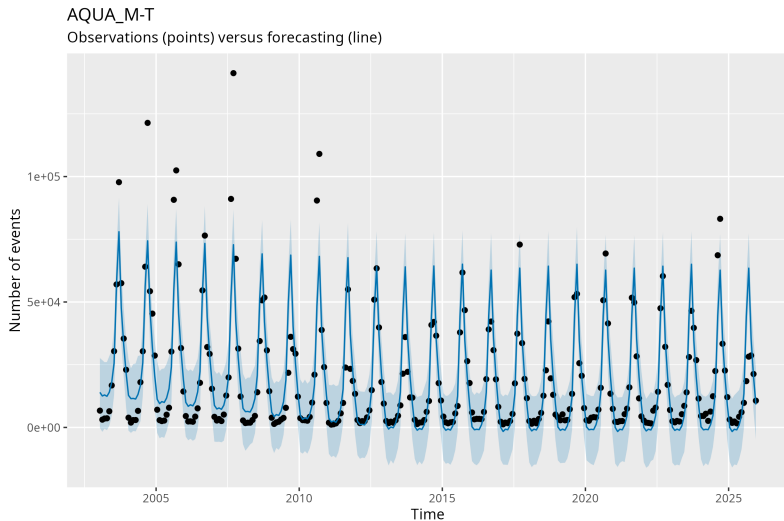
Time series seasonality

AQUA_M-T

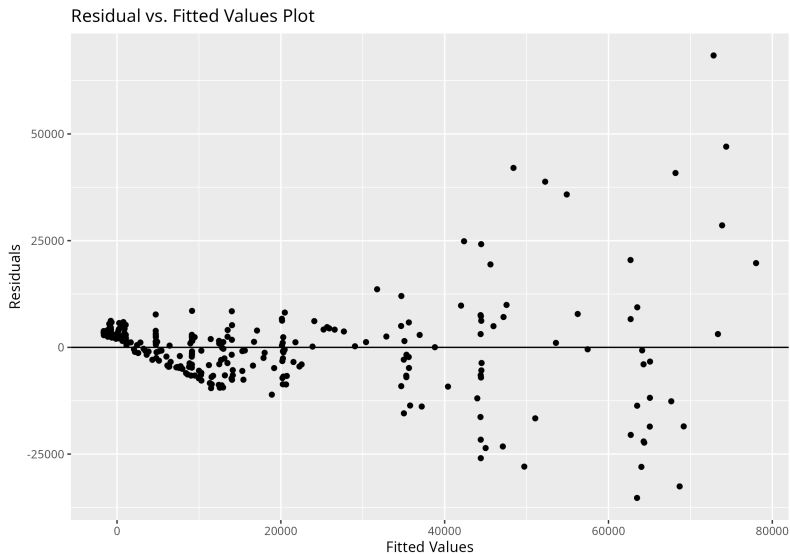
Do the data have a seasonality component?



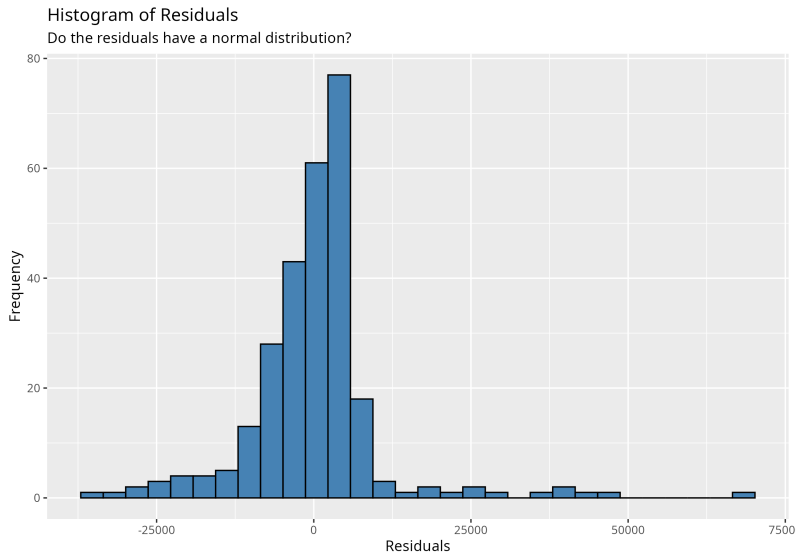
Time series versus forecast



Time series versus forecast



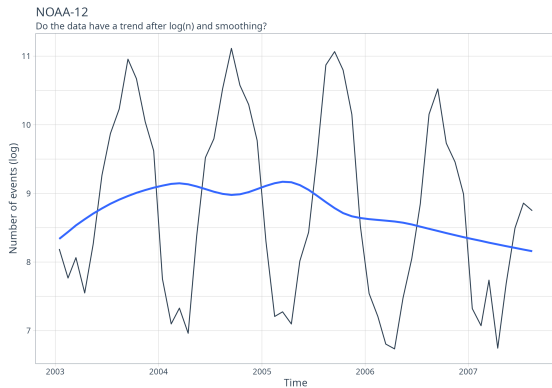
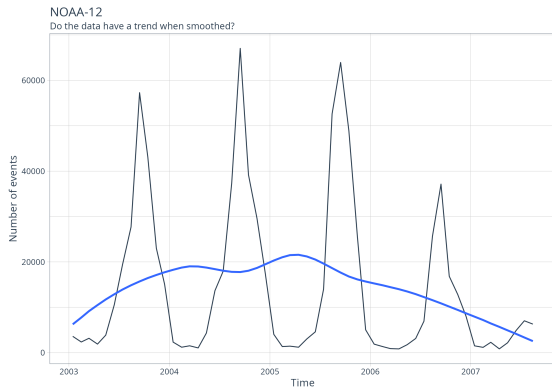
Time series versus forecast



NOAA 12



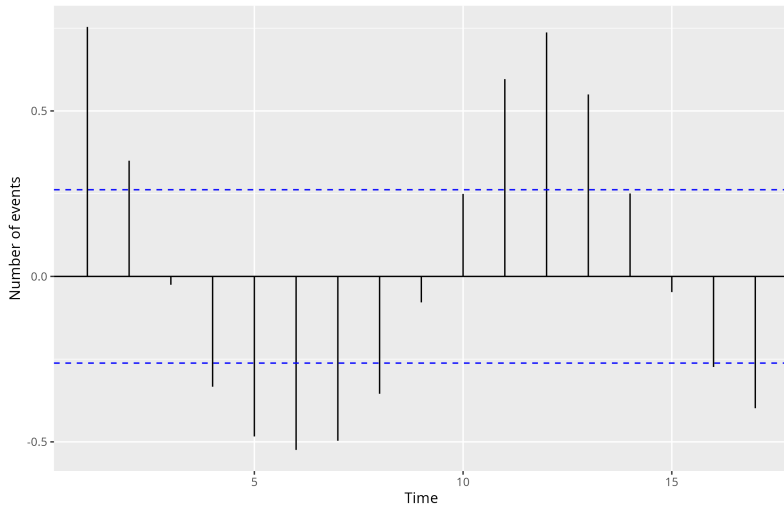
Time series trend (with smoothing)



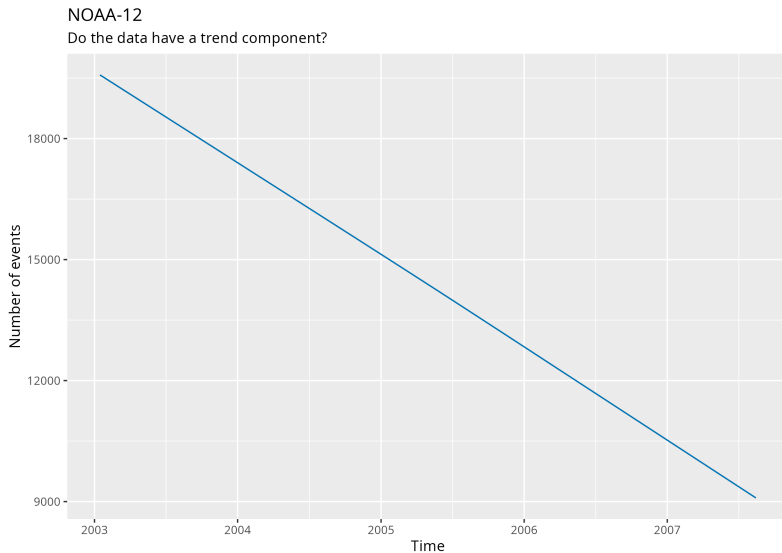
Time series autocorrelation

NOAA-12

Is the time series autocorrelated?



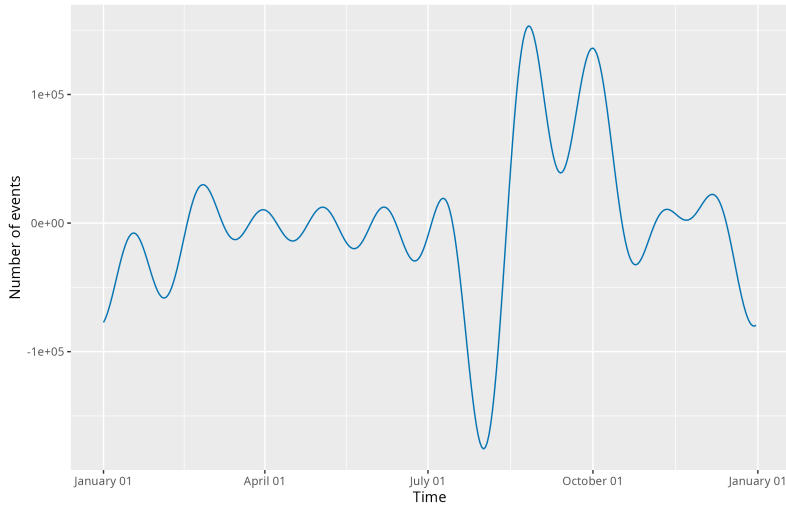
Time series trend



Time series seasonality

NOAA-12

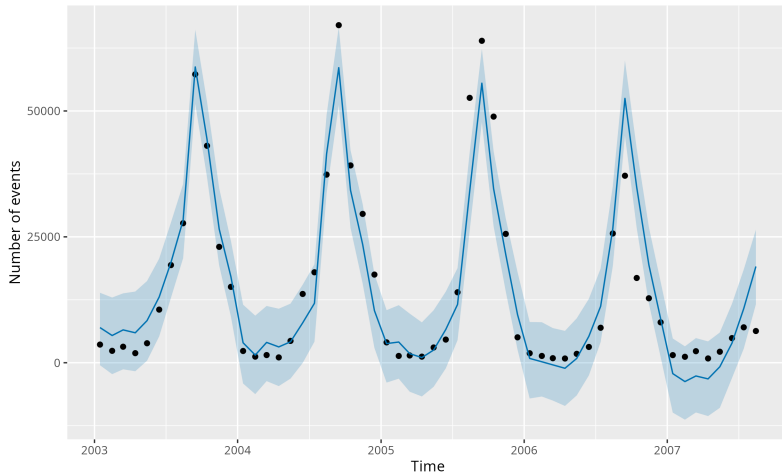
Do the data have a seasonality component?



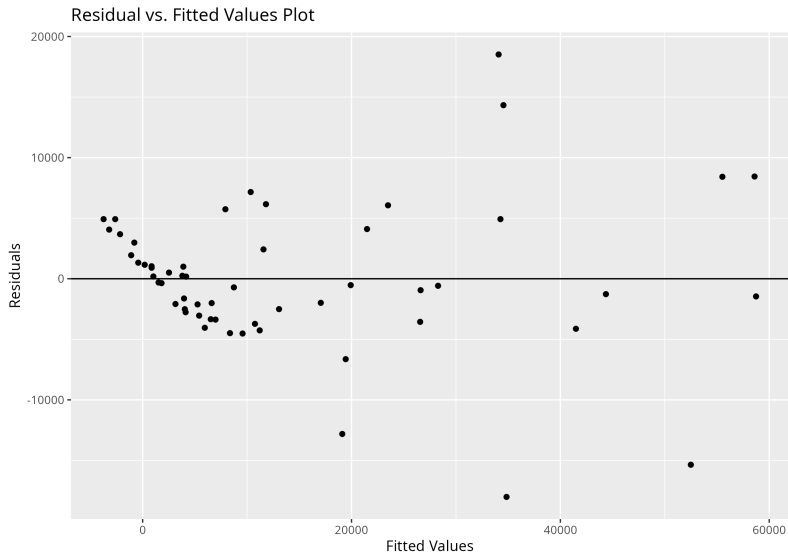
Time series versus forecast

NOAA-12

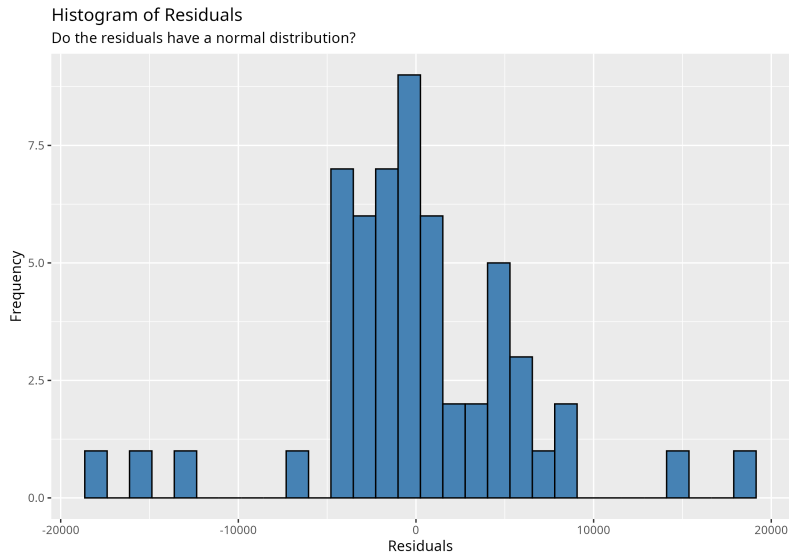
Observations (points) versus forecasting (line)



Time series versus forecast



Time series versus forecast



Summary

Name	Trend	TLog	Autocor	Season	Random Res	Res Dist
NPP-375D	+	+	12 m	yes	TODO	Normal
NPP-375-PM	+	+	12 m	yes	TODO	Normal
AQUA M T	-+	-+	12 m	yes	TODO	Normal
NOAA 12	-	+-	12 m	yes	TODO	Normal

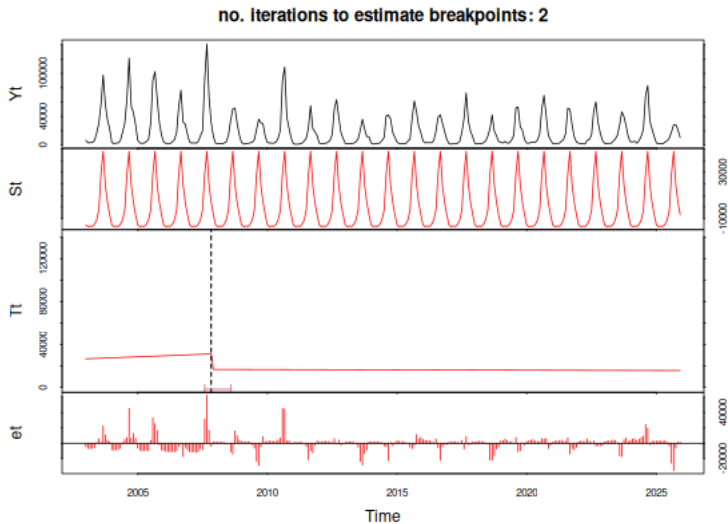
BFast

Breaks For Additive Seasonal and Trend - BFAST

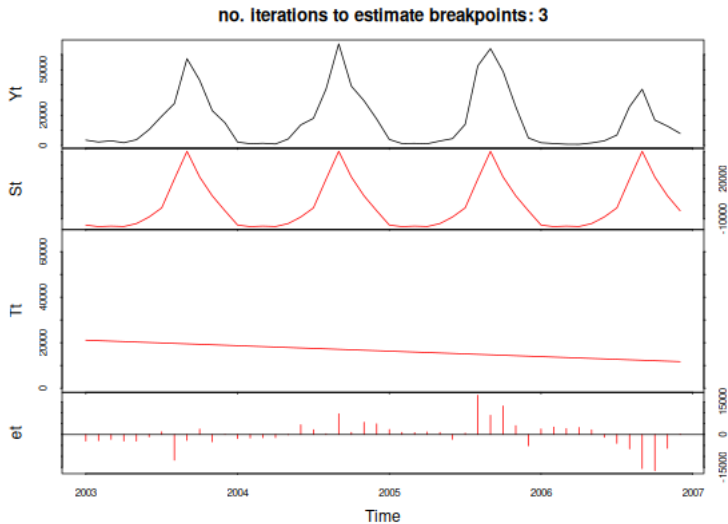
- Additive decomposition model that iteratively fits a linear trend and a seasonal model.
- Y_t is the observed data at time t .
- T_t is the trend component.
- S_t is the season component.
- e_t is the remaining variation in the data beyond the seasonal and trend components.
- [5].

$$Y_t = T_t + S_t + e_t$$

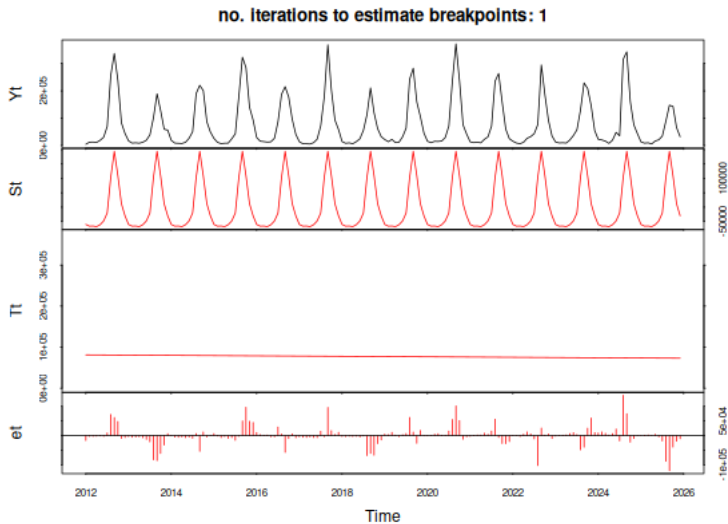
Time series Breaks (BFast) AQUA M-T



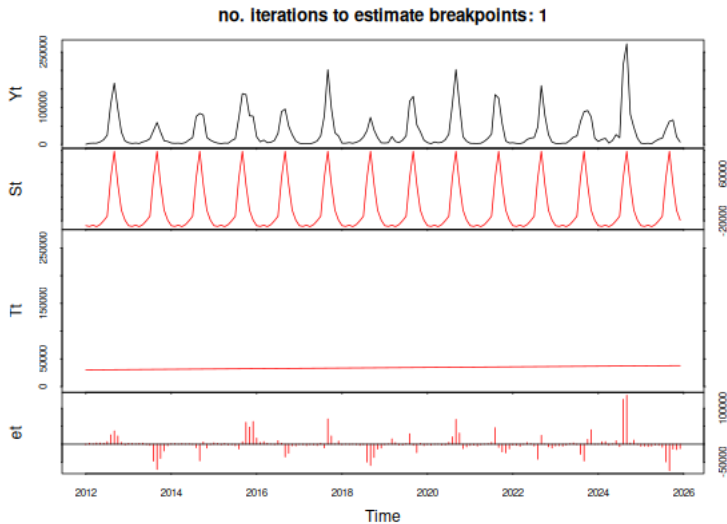
Time series Breaks (BFast) NOAA-12



Time series Breaks (BFast) NPP-375-PM



Time series Breaks (BFast) NPP-375-PM



Queimadas method (1-month regression)



Queimadas (1-month) method for forecasting heat foci

- 1 Get time series of two reference satellites:
 - ▶ One which we would like to extend in time (A.K.A. y).
 - ▶ The other one covering the whole time line we would like to forecast (A.K.A. x).
- 2 Adjust a linear model ($y = Ax + b$), in a month-by-month basis, to the overlapping observations in the time series of two reference satellites, using only the month of the observations.
- 3 Use the models to predict the whole time line.
- 4 Check forecast results.

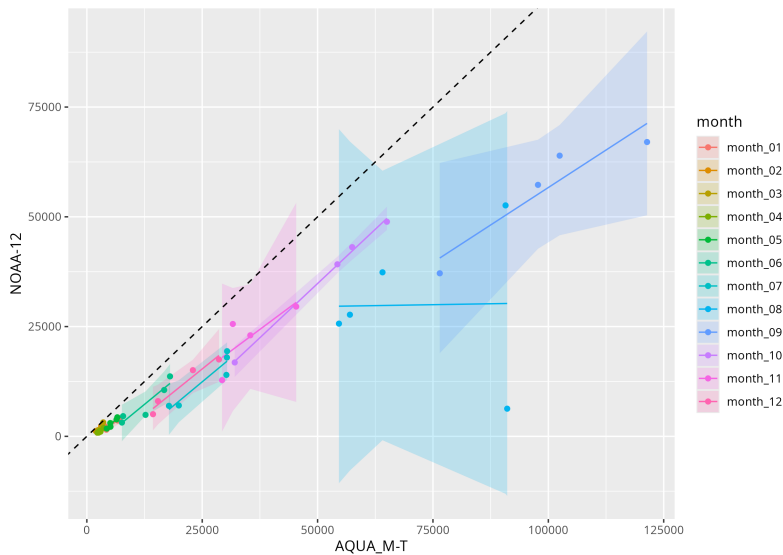


Confidence intervals

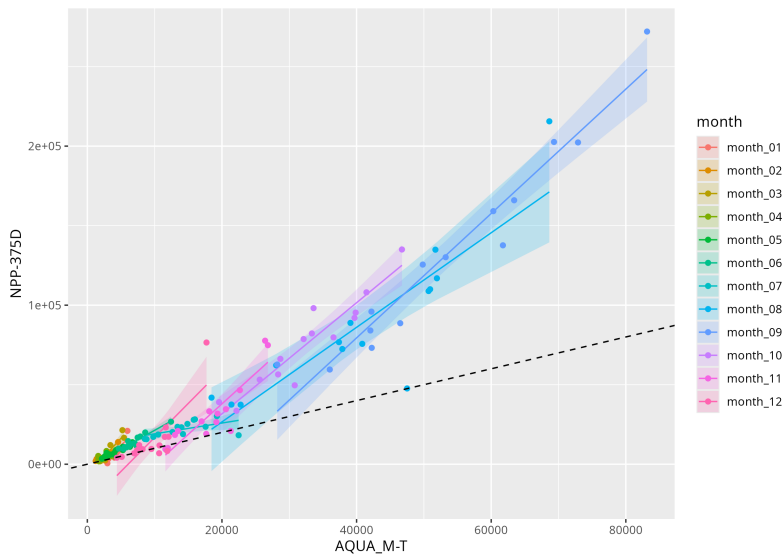
- The confidence intervals (gray shadows in the regression and forecast figures) is given at the 95% confidence level.



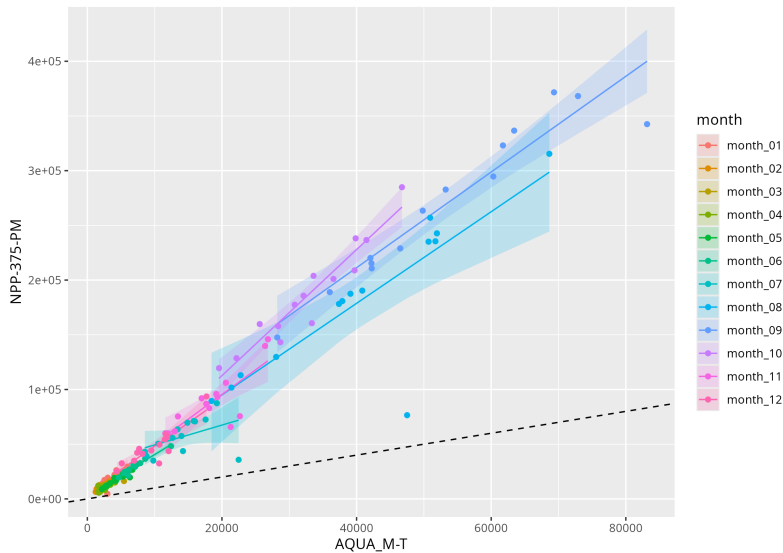
Linear model NOAA-12 vs. AQUA_M-T



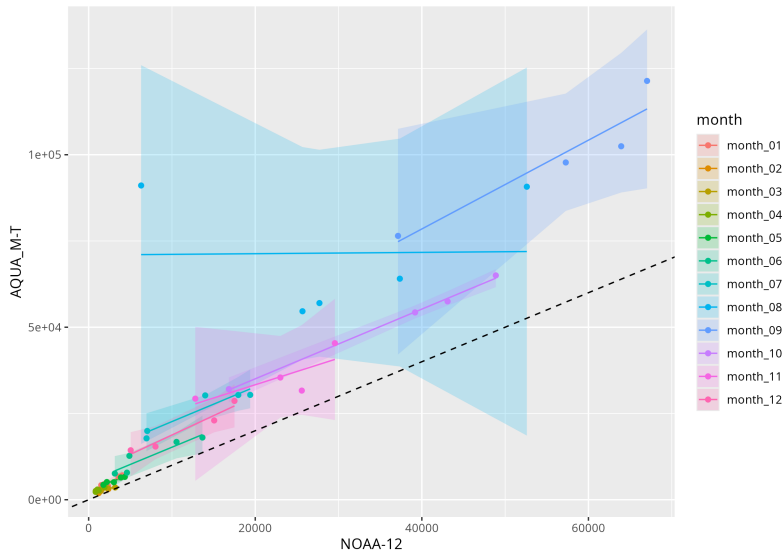
Linear model NPP-375D vs. AQUA_M-T



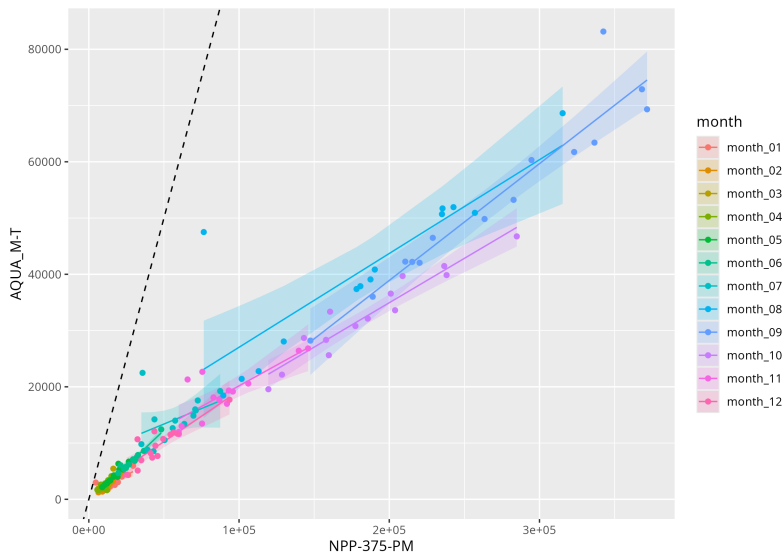
Linear model NPP-375-PM vs. AQUA_M-T



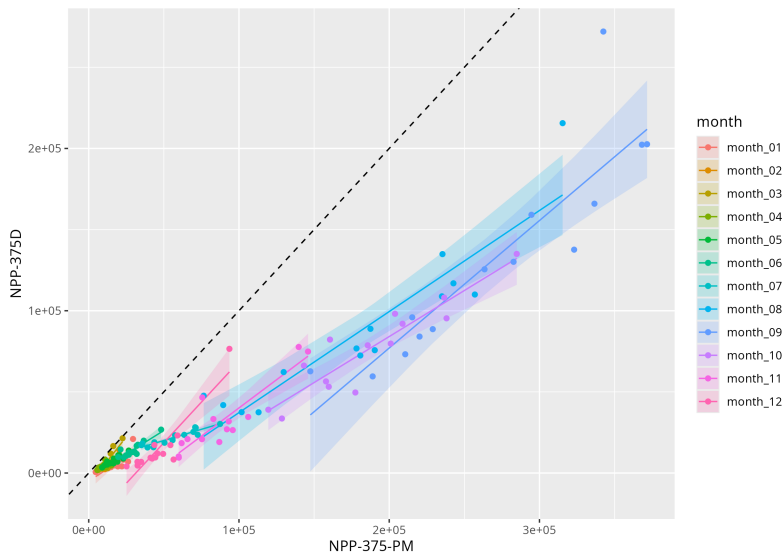
Linear model AQUA_M-T vs. NOAA-12



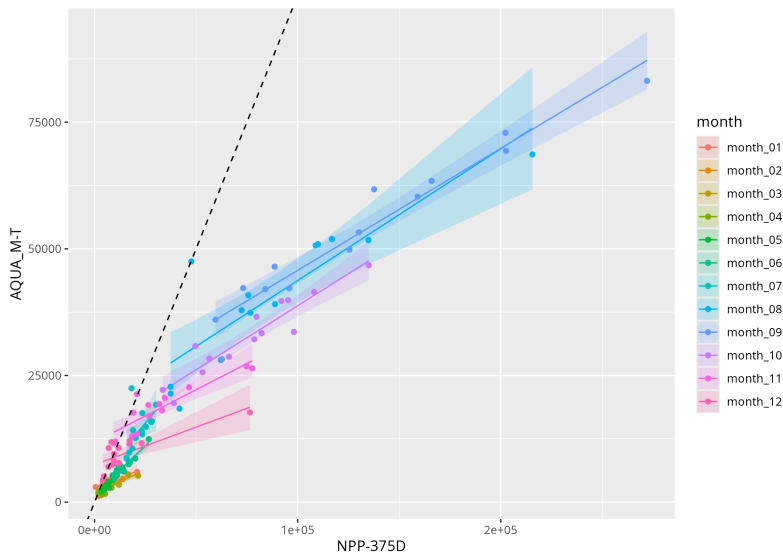
Linear model AQUA_M-T vs. NPP-375-PM



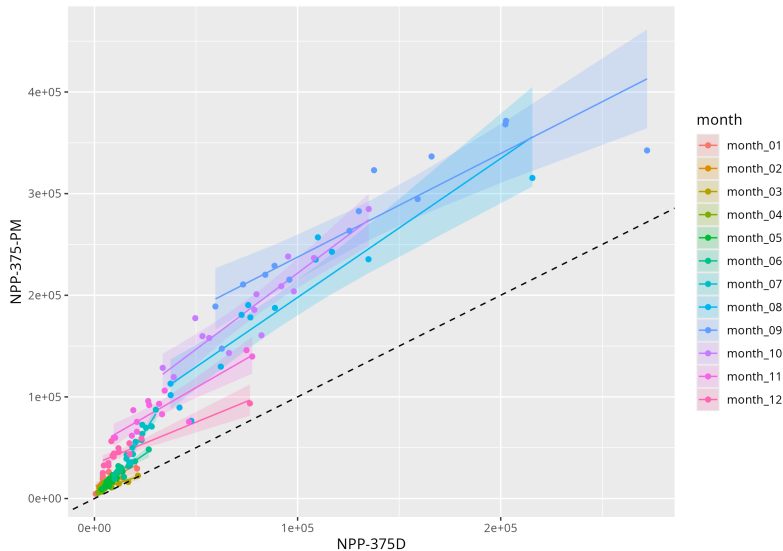
Linear model NPP-375D vs. NPP-375-PM



Linear model AQUA_M-T vs. NPP-375D



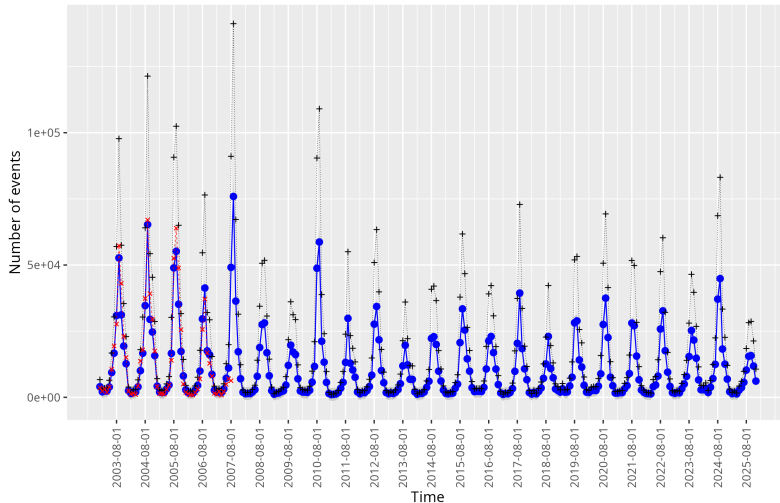
Linear model NPP-375-PM vs. NPP-375D



Stretch NOAA-12 to match AQUA_M-T (month-by-month)

Transformation of AQUA_M-T into NOAA-12

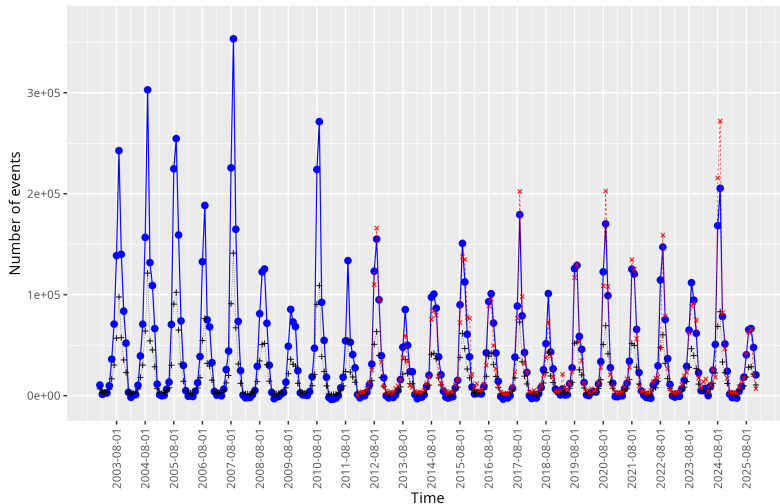
Forecast (blue line) versus observations (NOAA-12 red crosses, AQUA_M-T black plusses)



Stretch NPP-375D to match AQUA_M-T (month-by-month)

Transformation of AQUA_M-T into NPP-375D

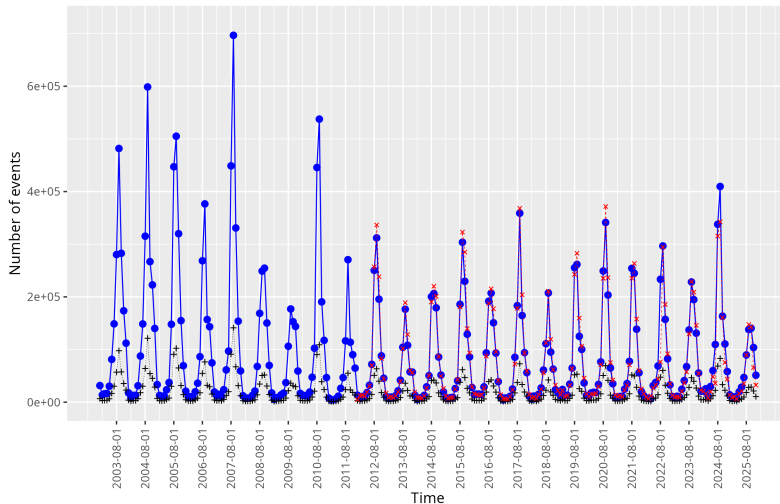
Forecast (blue line) versus observations (NPP-375D red crosses, AQUA_M-T black plusses)



Stretch NPP-375-PM to match AQUA_M-T (month-by-month)

Transformation of AQUA_M-T into NPP-375-PM

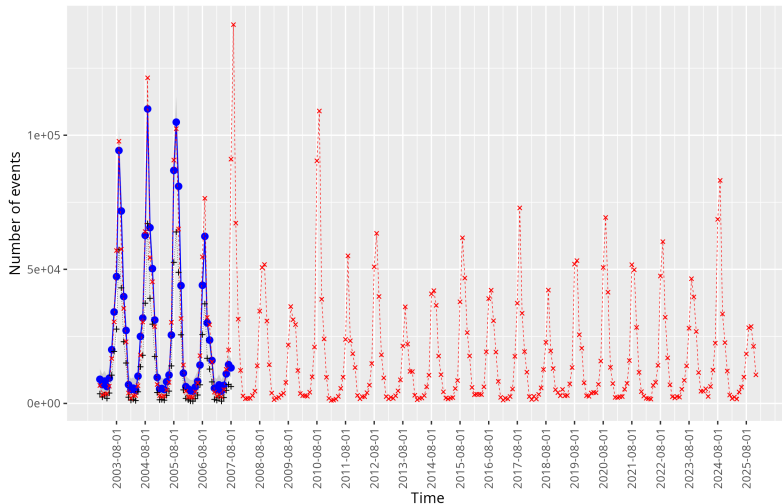
Forecast (blue line) versus observations (NPP-375-PM red crosses, AQUA_M-T black plusses)



Stretch match AQUA_M-T to match NOAA-12 (month-by-month)

Transformation of NOAA-12 into AQUA_M-T

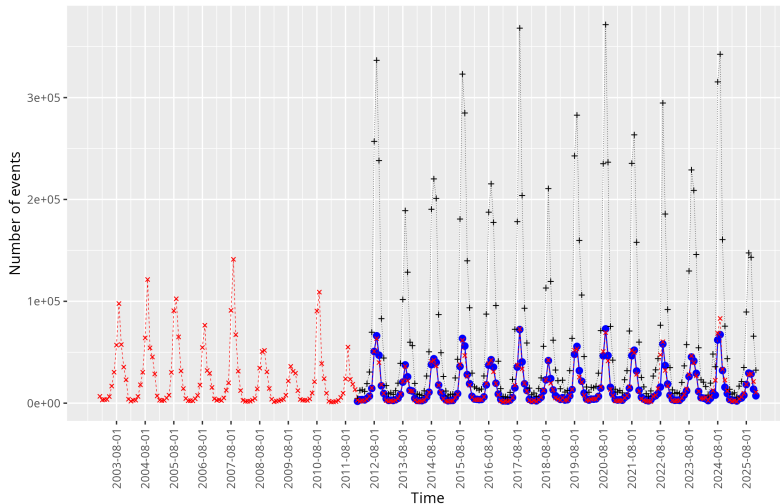
Forecast (blue line) versus observations (AQUA_M-T red crosses, NOAA-12 black plusses)



Stretch match AQUA_M-T to match NPP-375-PM (month-by-month)

Transformation of NPP-375-PM into AQUA_M-T

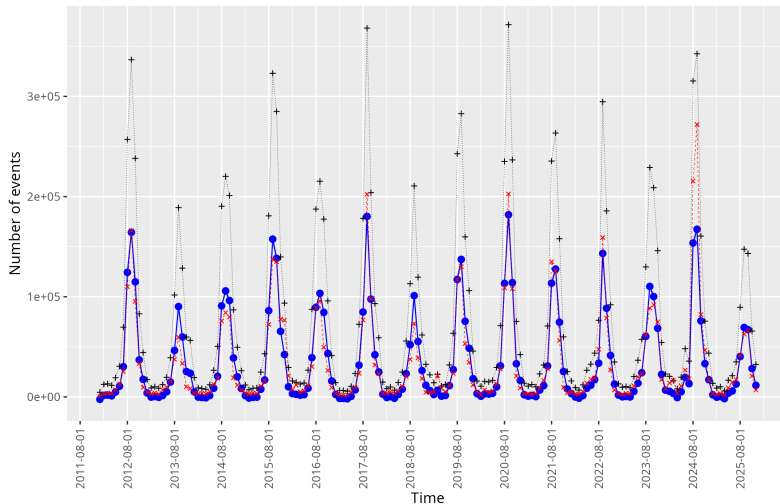
Forecast (blue line) versus observations (AQUA_M-T red crosses, NPP-375-PM black plusses)



Stretch match NPP-375D to match NPP-375-PM (month-by-month)

Transformation of NPP-375-PM into NPP-375D

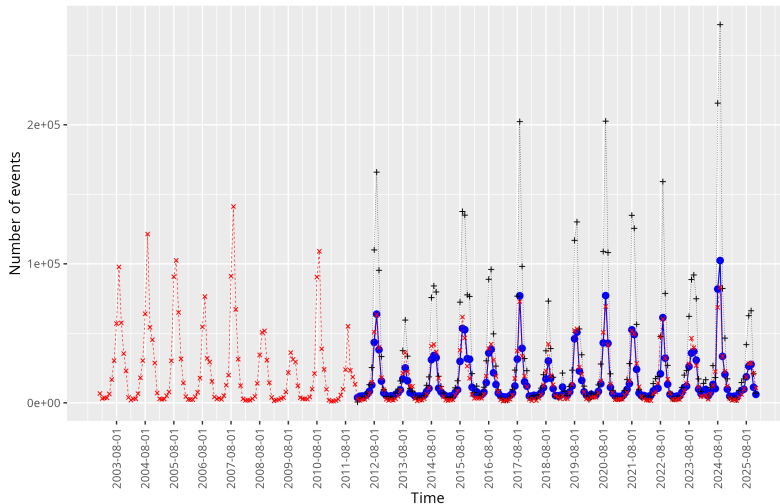
Forecast (blue line) versus observations (NPP-375D red crosses, NPP-375-PM black plusses)



Stretch match AQUA_M-T to match NPP-375D (month-by-month)

Transformation of NPP-375D into AQUA_M-T

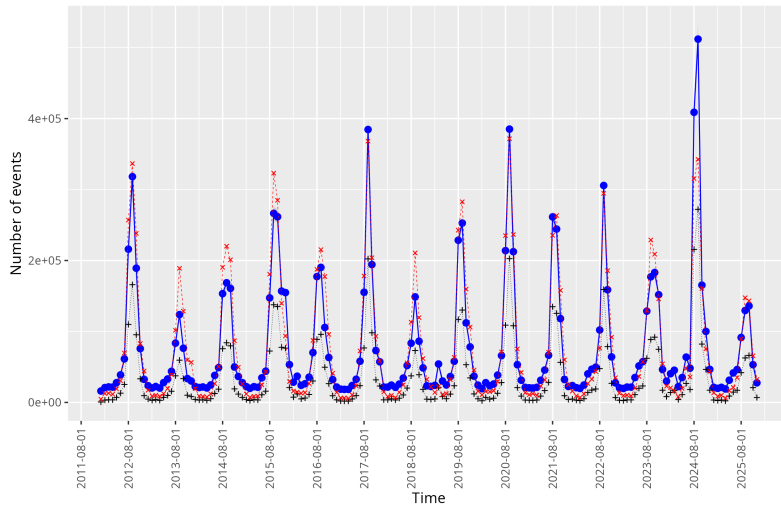
Forecast (blue line) versus observations (AQUA_M-T red crosses, NPP-375D black plusses)



Stretch match NPP-375-PM to match NPP-375D (month-by-month)

Transformation of NPP-375D into NPP-375-PM

Forecast (blue line) versus observations (NPP-375-PM red crosses, NPP-375D black plusses)



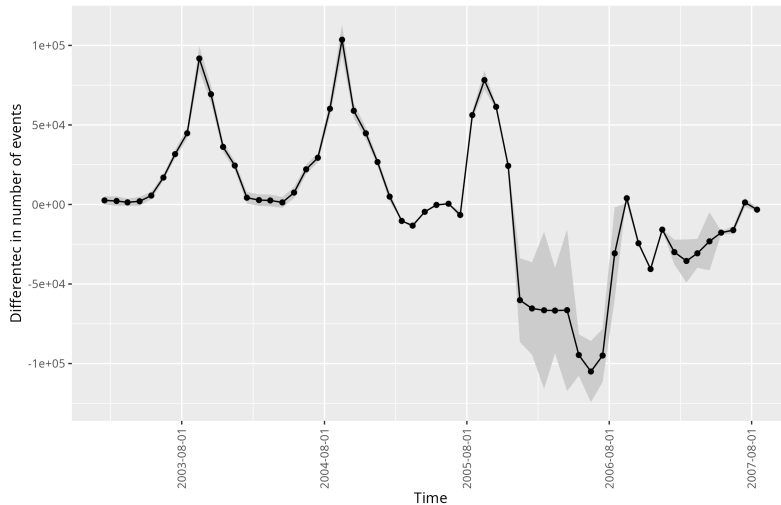
Forecast differences between Queimadas' 12 versus 1 month method



Forecast differences NOAA-12 vs. AQUA_M-T

Forecast differences using the Queimadas method 12 versus 1 month

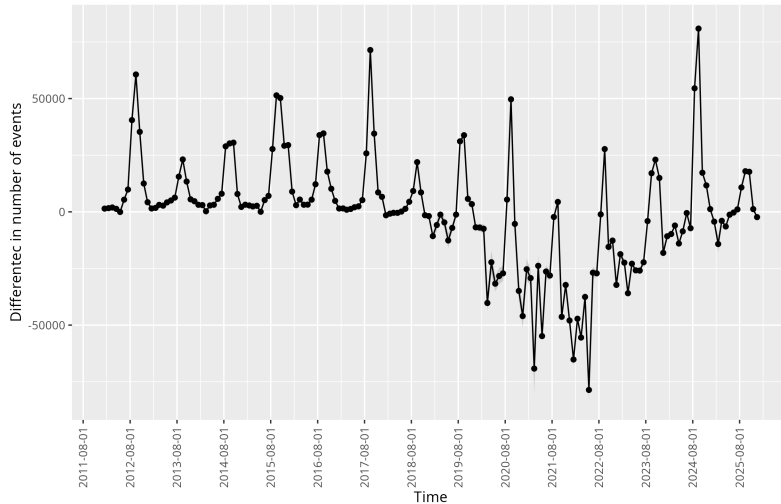
Differences in forecasts of AQUA_M-T using NOAA-12



Forecast differences NPP-375D vs. AQUA_M-T

Forecast differences using the Queimadas method 12 versus 1 month

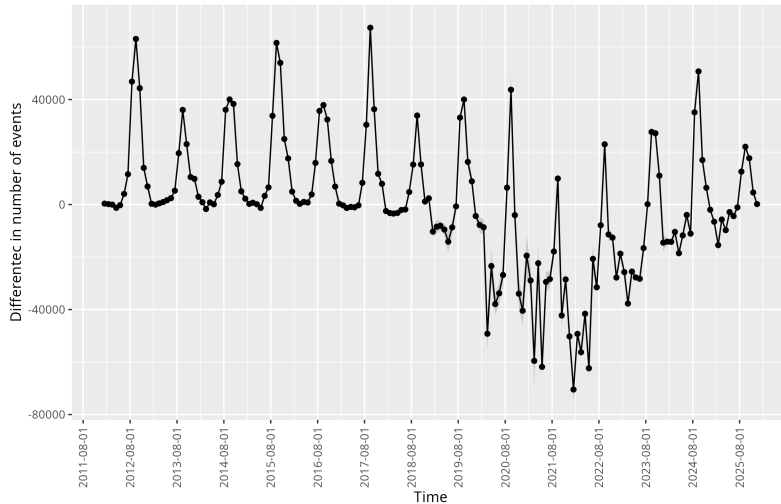
Differences in forecasts of AQUA_M-T using NPP-375D



Forecast differences NPP-375PM vs. AQUA_M-T

Forecast differences using the Queimadas method 12 versus 1 month

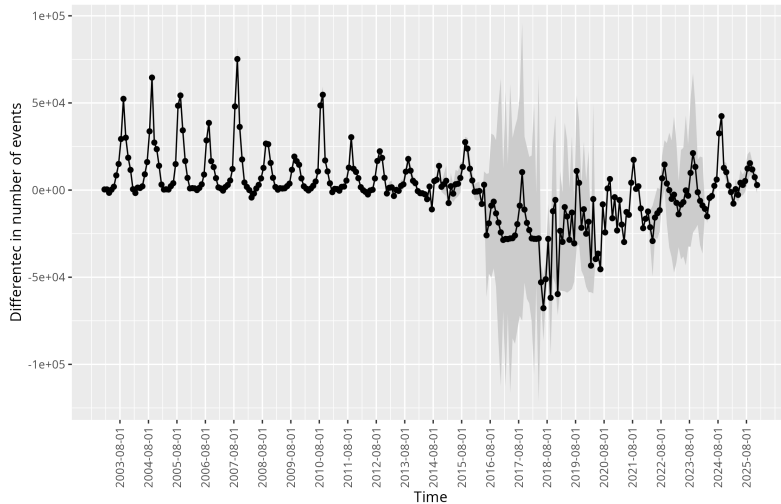
Differences in forecasts of AQUA_M-T using NPP-375-PM



Forecast differences AQUA_M-T vs. NOAA-12

Forecast differences using the Queimadas method 12 versus 1 month

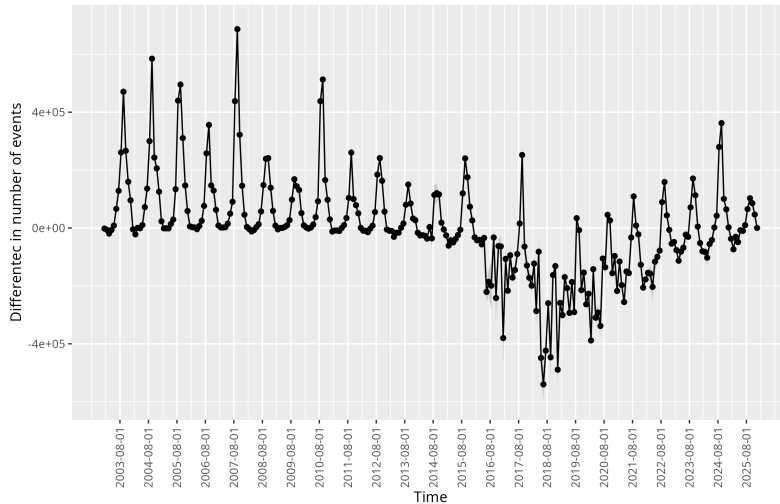
Differences in forecasts of NOAA-12 using AQUA_M-T



Forecast differences AQUA_M-T vs. NPP-375PM

Forecast differences using the Queimadas method 12 versus 1 month

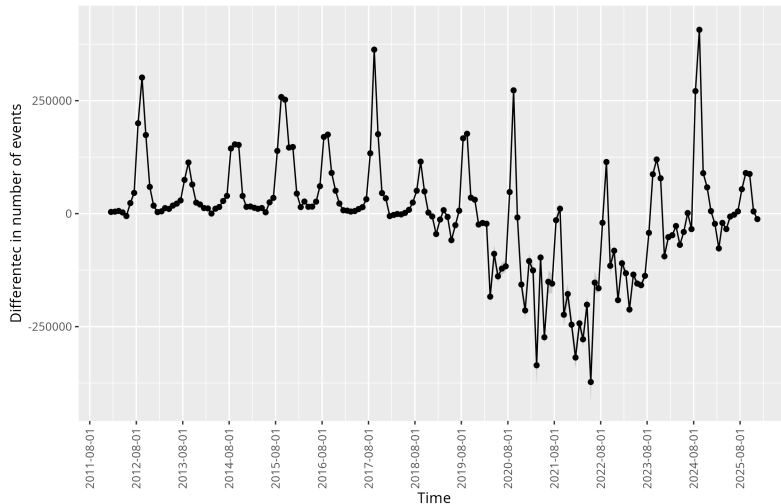
Differences in forecasts of NPP-375-PM using AQUA_M-T



Forecast differences NPP-375D vs. NPP-375PM

Forecast differences using the Queimadas method 12 versus 1 month

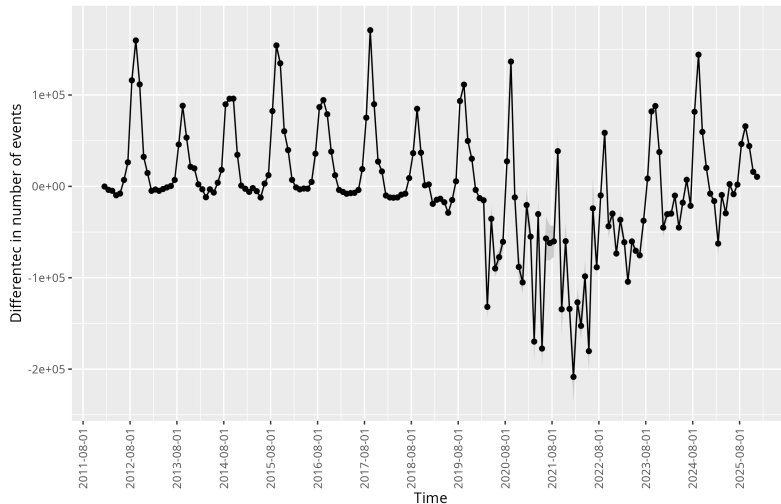
Differences in forecasts of NPP-375-PM using NPP-375D



Forecast differences NPP-375PM vs. NPP-375D

Forecast differences using the Queimadas method 12 versus 1 month

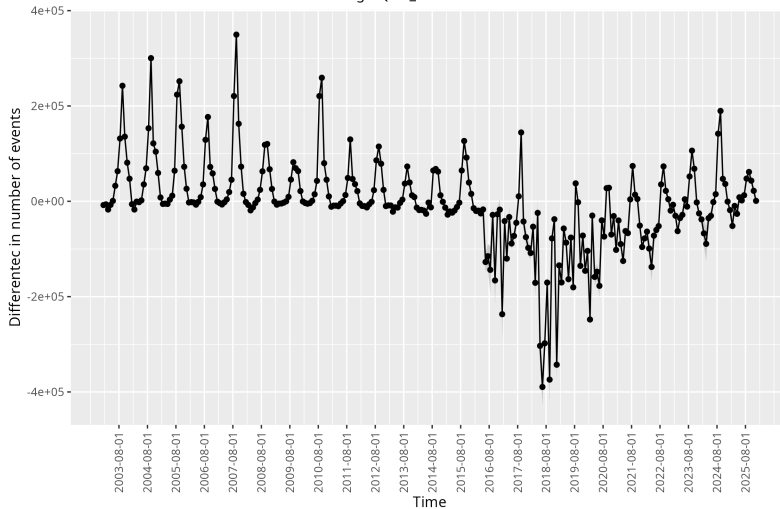
Differences in forecasts of NPP-375D using NPP-375-PM



Forecast differences AQUA_M-T vs. NPP-375D

Forecast differences using the Queimadas method 12 versus 1 month

Differences in forecasts of NPP-375D using AQUA_M-T



Future Work



Future work

- Use a curve instead of a line for regressions. NOTE: This could have been superseded by using month-by-month regressions.
- Get data from before 2003. It's not available at Queimadas' portal:
https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anual/Brasil_todos_sats
- Use Landsat grid for spatial aggregation.
- Try segmented regression on time series:
 - ▶ Use BFast to identify breakpoints and adjust models to each segment [6].
 - ▶ Use Mann-Kendall to identify breakpoints and adjust models to each segment. The *segmented* package has an implementation ??.



References



References I

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- [3] Alberto Setzer, Fabiano Morelli, and Jean Carlos Souza. “O Banco de Dados de Queimadas Do INPE”. In: *Biodiversidade Brasileira* 9.1 (2019), pp. 239–239.
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References II

- [6] Jan Verbesselt et al. “Phenological Change Detection While Accounting for Abrupt and Gradual Trends in Satellite Image Time Series”. In: *Remote Sensing of Environment* 114.12 (Dec. 2010), pp. 2970–2980. ISSN: 00344257. DOI: [10.1016/j.rse.2010.08.003](https://doi.org/10.1016/j.rse.2010.08.003). PMID: 11818969.

